

JEE MAIN REVISION MASTER: ELECTROSTATICS

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 53 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Coulomb's Law, Electric Field & Dipoles	35 PYQs	Electrostatic Forces
Very High	Gauss's Law & Electric Potential	24 PYQs	Flux & Potential

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Coulomb's Law, Electric Field & Dipoles	$E_{\text{axial}} = \frac{1}{4\pi\epsilon_0} \frac{2p}{r^3}$, $E_{\text{eq}} = \frac{1}{4\pi\epsilon_0} \frac{p}{r^3}$, $\tau = pE \sin\theta$, $W = pE(\cos\theta_1 - \cos\theta_2)$
Gauss's Law & Electric Potential	$\Phi = \frac{q_{\text{in}}}{\epsilon_0}$, $E_{\text{sheet}} = \frac{\sigma}{2\epsilon_0}$, $E = -\frac{dV}{dr}$, $V_{\text{inside sphere}} = \frac{q}{4\pi\epsilon_0 R}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Coulomb's Law, Electric Field & Dipoles (Very High Yield — 35 PYQs)

Core Principle: Coulomb's Law: $F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$. Electric dipole $\vec{p} = q(2\vec{a})$. Field at axial point $E_{\text{axial}} = \frac{2kp}{r^3}$; field at equatorial point $E_{\text{eq}} = \frac{kp}{r^3} = \frac{1}{2}E_{\text{axial}}$. Torque on dipole $\vec{\tau} = \vec{p} \times \vec{E}$; potential energy $U = -\vec{p} \cdot \vec{E} = -pE \cos\theta$.

■ **Caution / JEE Pitfall:** Field of a short dipole falls off as $1/r^3$, NOT $1/r^2$ like a point charge!

■ **Speed Trick:** Work done to rotate a dipole from stable equilibrium ($\theta = 0^\circ$) to unstable equilibrium ($\theta = 180^\circ$) is $W = 2pE$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "An electron falls through a distance 1.5 cm in 2.0×10^4 N/C from rest. The time taken to cover this distance in second is ____ ..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "An electric dipole of dipole moment $\vec{p}$ is placed in the uniform electric field $\vec{E}$. Then which of the following statements are..." [Ans: N/A]
- [2025 Shift] Q#1 (MCQ): "Charges are uniformly spread on the surface of a conducting sphere. The electric field from the centre of sphere to a point outside the sphere..." [Ans: N/A]

3.2 Gauss's Law & Electric Potential (Very High Yield — 24 PYQs)

Core Principle: Gauss's law $\Phi = \frac{q_{\text{enc}}}{\epsilon_0}$. Applications: Line charge $E = \frac{\lambda}{2\pi\epsilon_0 r}$, infinite non-conducting sheet $E = \frac{\sigma}{2\epsilon_0}$ (independent of distance). Conducting sphere: $E_{\text{(in)}} = 0$, $V_{\text{(in)}} = V_{\text{(surface)}} = \frac{kq}{R}$. Equipotential surfaces are always perpendicular to electric field lines; work done moving a charge on an equipotential surface is zero.

■ **Caution / JEE Pitfall:** Electric potential inside a charged conducting hollow sphere is constant and non-zero ($V = \frac{kq}{R}$), while electric field inside is zero!

■ **Speed Trick:** If n identical mercury droplets each of potential V coalesce to form one big drop, the new potential is $V' = n^{2/3} V$.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "A 200 J of work is done in moving a charge 5 C from a point A where the potential is -20 V to another point B where potential is V volt..." **[Ans: N/A]**
- **[2026 Shift] Q#3 (MCQ):** "Consider three point charges $-2Q, Q$ and $-Q$ and three surfaces $S_{(1)}, S_{(2)}$ and $S_{(3)}$ as shown in the figure. Match the entries of List..." **[Ans: N/A]**
- **[2026 Shift] Q#4 (MCQ):** "What will be the total electric flux through the faces of the cube as given in the figure with side of length 'a' if a charge Q is placed..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

A 200 J of work is done in moving a charge 5 C from a point A where the potential is -20 V to another point B where potential is V volt. The value of V at B is

- (A) 10 V
- (B) 20 V
- (C) 40 V
- (D) 60 V

Correct Answer: None

Solution Analysis: Work done in moving a charge from one point to another is related to potential difference by $W = q(V_B - V_A)$. Here, $W = 200\text{ J}$, $q = 5\text{ C}$, $V_A = -20\text{ V}$. So, $200 = 5[V_B - (-20)]$ $200 = 5(V_B + 20)$. Divide both sides by 5 : $40 = V_B + 20$ $V_B = 20\text{ V}$. So, the value of potential at point B is 20V. Correct option: Option B.

Benchmark Problem 2: KCET 2026 (Q#1)

An electron falls through a distance 1.5 cm in 2.0×10^4 N/C from rest. The time taken to cover this distance in second is ____ (e = 1.6×10^{-19} C, $m_e = 9.11 \times 10^{-31}$ kg)

- (A) 2.9×10^{-9}
- (B) 2.9×10^9
- (C) 4×10^{-6}
- (D) 4×10^6

Correct Answer: None

Solution Analysis: The electron starts from rest, so its initial velocity is $u=0$. The electric force on the electron is $F=eE$. Using Newton's second law, $F=ma$ so the acceleration is $a=\frac{eE}{m_e}$. Substitute the given values: $a=\frac{(1.6 \times 10^{-19})(2.0 \times 10^4)}{9.11 \times 10^{-31}}$. $a=\frac{3.2 \times 10^{-15}}{9.11 \times 10^{-31}}$. $a \approx 3.51 \times 10^{15} \text{ m/s}^2$. The distance covered is $s=1.5 \text{ cm}=1.5 \times 10^{-2} \text{ m}$. Since the electron starts from rest, $s=\frac{1}{2}at^2$. So, $t=\sqrt{\frac{2s}{a}}$. $t=\sqrt{\frac{2(1.5 \times 10^{-2})}{3.51 \times 10^{15}}}$.